Ethical Web Development Principles

This MERN project, The Village, will be developed with a commitment to ethical web development. We will design our application to be accessible, inclusive, and usable for all users through progressive enhancement, responsive design, and performance-conscious implementation. We prioritise privacy and security by enforcing HTTPS, safeguarding user data, respecting tracking preferences, and providing transparency around data usage and export.

We also uphold professional responsibility by maintaining well-documented, tested, and version-controlled code in Git/ GitHub, and by engaging respectfully with the wider development community. These principles guide our technical and architectural decisions throughout the project.

Web applications should work for everyone

Progressive enhancement means building the application so that its core functionality works first, and then layering more advanced features on top for browsers and devices that can support them.

This means basic content and functionality (forms, navigation, data display) should work even if the JavaScript fails or loads slowly. React interactivity enhances the experience, but the user should not be completely blocked without it and the pages should render meaningful content quickly before advanced UI behaviours initialise. Accessibility and semantic HTML come first; styling and dynamic features are added afterward. And finally, features that rely on modern APIs should degrade gracefully rather than break.

The Village prioritises accessibility, which means designing the health and wellness app so it can be safely and comfortably used by as many people as possible, including those with disabilities, temporary impairments, or heightened emotional sensitivity.

In practice, this means using semantic HTML so screen readers can accurately interpret content and ensuring full keyboard navigation for users who cannot use a mouse. Providing gentle colour schemes and readable typography to reduce visual strain as well as writing clear, simple language to support users experiencing stress, anxiety, or cognitive overload, and avoiding flashing elements or overwhelming UI patterns that could negatively impact vulnerable users.

In The Village, developing inclusive forms means designing input experiences that are clear, supportive, and non-judgemental, particularly given the sensitivity of mental health and pain information. Forms should use plain, empathetic language, clearly explain why information is being collected, and avoid assumptions about identity, relationships, or lived experience.

Validation feedback must be constructive and easy to understand, guiding users without creating frustration or shame. Input fields should support assistive technologies, allow flexible responses where appropriate, and minimise cognitive load so that users can complete tasks comfortably, even during moments of stress or emotional vulnerability.

Web applications should work everywhere

In The Village, responsive design ensures the interface adapts seamlessly across mobile phones, tablets, and desktops. Users seeking mental health and pain support may access the platform from a variety of environments, so layouts, typography, and interactions must remain clear, stable, and usable regardless of screen size or device capability. Touch targets, spacing, and text scaling are carefully considered to maintain comfort and readability. This approach ensures that the experience feels consistent and supportive, whether accessed privately at home or on the move.

Performance is prioritised to reduce load times and friction, particularly for users on slower or unstable connections. Efficient rendering, React lazy loading, optimised assets, and thoughtful data handling help ensure that users can access support quickly, without unnecessary technical barriers. In a wellbeing context, delays and unresponsive interfaces can increase frustration or distress, so speed and reliability directly contribute to user trust. Monitoring and optimisation are treated as ongoing responsibilities rather than one-time tasks.

Where appropriate, offline-first functionality with service-workers allows essential features to remain available during temporary connectivity loss. This improves reliability and ensures users can still access important information or continue interactions even when internet access is interrupted. Caching critical resources and preserving in-progress data reduces the risk of losing meaningful input. This resilience is especially important when users are engaging with personal reflections or sensitive material.

Clear, permanent, human-readable URLs enable users to bookmark, revisit, and share specific resources with confidence. Deep linking with React Router also supports accessibility and transparency by ensuring content can be directly accessed without relying solely on client-side navigation. Structured routing improves search visibility and makes the platform easier to understand conceptually. It also allows support resources to be referenced reliably in external communications, strengthening continuity of care and access.

Web applications should respect a user's privacy and security

In The Village, all communication between the client and server is secured using HTTPS, TLS certificates and Express enforcement to encrypt data in transit. This protects sensitive mental health and physical pain information from interception or tampering during transmission. Enforcing secure connections across all routes, APIs, and assets reinforces user trust and ensures that privacy is treated as a non-negotiable foundation rather than an optional feature.

User tracking preferences are honoured by minimising unnecessary data collection and providing clear choices around analytics or cookies. Where tracking is used to improve functionality, it is implemented transparently and only with informed consent. Respecting these preferences acknowledges user autonomy and recognises that individuals engaging with mental health and pain tracking services may have heightened concerns about digital privacy.

Users are given clear, accessible explanations about what information is collected, why it is needed, and how it is used. Privacy policies and consent prompts are written in plain language rather than legal or technical jargon. This transparency empowers users to make informed decisions and reduces uncertainty, which is particularly important in a wellbeing-focused environment.

User data is protected through secure storage practices, including encryption, access controls, and proper authentication mechanisms with JWTs and MongoDB Atlas. Sensitive information such as identity, mental health and pain information is only accessible to authorised systems and is handled according to the principle of least privilege. Regular review of security measures helps ensure that protections evolve alongside emerging risks, maintaining a safe and trustworthy platform for all users.

Web developers should be considerate of their peers

In The Village, we maintain clear inline comments and structured documentation to support maintainability and collaboration. Code is documented using meaningful function descriptions and README files within our GitHub repository, ensuring other developers can understand setup, architecture, and design decisions. Where appropriate, JSDoc-style comments are used in JavaScript to clarify parameters and return values.

To maintain code quality and consistency, we use ESLint to enforce coding standards across the MERN stack. Testing is conducted using tools such as Jest and React Testing Library to validate core functionality and reduce regressions. These tools help ensure that new changes do not unintentionally break existing features, supporting reliability and team accountability.

Version control is managed through Git and GitHub, enabling branching, commit history tracking, and collaborative development. Pull requests are used to review changes before merging into the main branch. Where possible, GitHub Actions can be configured to automatically run linting or tests on each push, supporting continuous integration and early error detection.

We recognise that the MERN stack relies heavily on open-source technologies, including React, Node.js, and MongoDB. When possible, we contribute by reporting issues, improving documentation, or following community guidelines. Professional respect is maintained through constructive communication in commits, pull requests, and discussions, fostering a positive and inclusive development environment.